

Trader Behavior vs Market Sentiment (Fear–Greed Index)

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Datasets Used

- 1 Bitcoin Market Sentiment Dataset (Fear & Greed Index)
- 2 Historical Trader Data from Hyperliquid

1. Objective

- This project explores the relationship between trader behavior and overall Bitcoin market sentiment.
- The goal is to understand whether trader performance and activity (PnL, volume, number of trades, trade size, win rate) aligns or diverges from sentiment conditions such as Fear, Neutral, and Greed.

2. Dataset Summary

A) Hyperliquid Historical Trader Data

- Rows: 211,224
- Columns: 16
- Key columns used: Timestamp IST (trade time), Size USD (trade size/volume), Closed PnL (profit/loss), Fee (transaction cost), Side, Coin, Trade ID

B) Bitcoin Market Sentiment (Fear–Greed Index)

- Rows: 2,644
- Columns: 4
- Key columns used: date, value (0–100 sentiment score), classification (Extreme Fear, Fear, Neutral, Greed, Extreme Greed)

3. Data Preprocessing & Feature Engineering

Step 1: Date-time handling

- Converted Timestamp IST to datetime format using dayfirst=True
- Extracted daily date as trade_date
- Converted sentiment date into datetime and extracted daily date

Step 2: Feature creation

- Created additional features: `volume_usd = abs(Size USD)`, `is_profit = Closed PnL > 0`, `abs_pnl = abs(Closed PnL)`

Step 3: Daily aggregation (Trader Behavior per day)

- Aggregated the trader dataset at daily level: `total_trades`, `total_volume_usd`, `avg_trade_size`, `total_pnl`, `avg_pnl`, `win_rate`, `total_fees`

Step 4: Merge

- Merged daily trader behavior with sentiment dataset on `trade_date`.
- Merged Dataset Shape: (158, 12) — 158 common days between both datasets.

4. Key Results & Analysis

Sentiment vs Trading Activity (Volume & Trades)

- Correlation with sentiment value: `value vs total_volume_usd = -0.188`; `value vs total_trades = -0.188`
- This indicates a weak negative relationship, meaning trading activity slightly increases during fear periods.
- Average activity by sentiment classification: Fear $\approx$ \$2.15M/day & 375 trades/day; Extreme Fear $\approx$ 388 trades/day; Extreme Greed $\approx$ \$0.42M/day
- **Insight:** Traders are most active during Fear, likely due to volatility spikes, panic exits, and increased market participation.

Sentiment vs Profitability (PnL)

- Correlation: `value vs total_pnl = -0.057` (almost no relationship).
- Profitability differs across sentiment classes: Fear $\approx$ \$48,087 (highest); Extreme Greed $\approx$ \$26,904; Greed $\approx$ \$10,886; Extreme Fear $\approx$ \$733 (lowest).
- **Insight:** Fear periods produced the highest profitability, while Extreme Fear is unstable and low-profit.

Win Rate vs Sentiment

- Correlation: `value vs win_rate = 0.006` (no relationship).
- Average win rate by sentiment: Greed 36.24% (highest); Fear 34.81%; Extreme Greed 31.96%; Extreme Fear 30.75%; Neutral 28.67% (lowest).
- **Insight:** Win rate is not controlled by sentiment. Traders win slightly more often in Greed, but that does not guarantee higher profits.

Trade Size (Risk Appetite) vs Sentiment

- Correlation: `value vs avg_trade_size = +0.171` (weak positive).
- Average trade size by sentiment: Extreme Greed \$7,954 (highest); Neutral \$7,192; Greed \$6,352; Fear \$6,015; Extreme Fear \$2,191 (lowest).

- **Insight:** During Extreme Greed, traders place larger trades, showing higher risk-taking behavior.

5. Hidden Trend / Signal

- **Major finding:** "More wins does not mean more profit"
- Greed has the highest win rate (36.24%), but Fear has the highest daily PnL (\$48,087).
- This suggests: In greed, traders may win small trades but miss large moves. In fear, volatility creates bigger opportunities, generating higher overall profits.

6. Recommendations (Smarter Trading Strategy)

- **1.** Focus on Fear periods (opportunity zone): Highest volume + highest daily PnL → Use disciplined entries and controlled risk.
- **2.** Avoid overconfidence in Extreme Greed: Highest trade size + lower win rate than Greed → Reduce leverage/position size, avoid FOMO trading.
- **3.** Be defensive in Extreme Fear: High trades but very low profit → Trade smaller, avoid revenge trading, wait for stability.
- **4.** Use combined signals: Fear + high volume → volatility opportunity signal; Extreme Greed + large trade size → FOMO risk signal.

7. Conclusion

- This analysis shows that trader behavior changes with market sentiment, especially in terms of risk appetite and participation.
- While sentiment does not directly predict win rate or PnL, Fear periods consistently show higher trading activity and profitability, whereas Greed periods show larger trade sizes but lower total profit.
- Combining sentiment with trading metrics can support more disciplined and smarter trading decisions.